DEYANG HSU

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EDUCATION

University of Southern California Los Angeles, California
Master's in Computer Science, Viterbi School of Engineering August 2024-May 2026

The Ohio State University Columbus, Ohio
Bachelor of Science in Electrical Computer Engineering August 2018-May 2022
Honors: Magna Cum Laude | Dean's List Scholar | Alumni Society Scholarship recipient GPA: 3.769/4.0

Relevant Coursework:
Machine Learning & AI | OOP | Discrete Structures | Data Structures and Algorithms |Computer Architecture | Parallel Computing | Database systems | Operating Systems | Web Crawling and Search Engines | LLM | NLP | Cybersecurity | Computer Networking | Multimodal Probabilistic Learning | Natural Language Dialogue Systems

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, HTML
Technologies & Tools: Text-to-LoRA, Doc-to-LoRA, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Pandas, NumPy, HuggingFace, NLTK, OpenMP, CUDA, SLURM, Multithreading, MPI, Relational Databases, Web Crawling, RAG
Systems: RISC-V, CISC, GitHub, Subversion, Networking, Operating Systems, Computer Architecture

RESEARCH EXPERIENCE

Research Assistant, FORTIS Lab | University of Southern California January 2025 – Present

- Conduct research within the FORTIS (Foundations Of Robust Trustworthy Intelligent Systems) Lab under **Professor Yue Zhao**, advancing research on robust and trustworthy AI systems across anomaly detection, out-of-distribution detection, graph learning, structured data modeling, and generative AI.
- Lead research on Comparative Narrative Analysis (CNA) across TELeR taxonomy levels 1 to 4, leveraging frontier LLMs (GPT-4, DeepSeek, Claude) to perform multi-narrative evaluation, conflict detection, and holistic summarization tasks.
- Developed and automated model evaluation pipelines for CNA tasks, improving assessment coverage across narrative conflict detection, overlap identification, uniqueness scoring, and holistic summarization.

PROJECTS

StressID: Physiological Audio Stress Detection Pipeline

- Designed and built an end-to-end audio ML pipeline for real-time stress detection from physiological audio signals, combining signal processing with deep learning classification models.
- Implemented acoustic feature extraction modules (MFCCs, pitch, prosodic features, voice activity detection), enabling precise discrimination of stress states from audio biomarkers.
- Fine-tuned pretrained speech models using LoRA adapters via HuggingFace, adapting large audio transformers for stress recognition with minimal computational overhead.

WORK EXPERIENCE

Intel Chandler, Arizona
Software Development Engineer June 2022-August 2023

- Diagnosed and resolved critical bugs across in-house software services, driving a 27% increase in system uptime and reducing production incidents
- Refactored and optimized internal production database systems with PL/SQL, significantly improving query performance, data reliability, and throughput
- Developed and deployed custom production software solutions, leading to an 18% rise in wafer production efficiency

General Motors Bay City, Michigan
Manufacturing Controls Engineer Intern June 2021-August 2021

- Diagnosed and resolved failures in robotics systems and PLCs, cutting production downtime by 8%
- Implemented an IO-Link pressure sensor system via Siemens SIMATIC STEP 7, reducing inventory scrapping by 13%